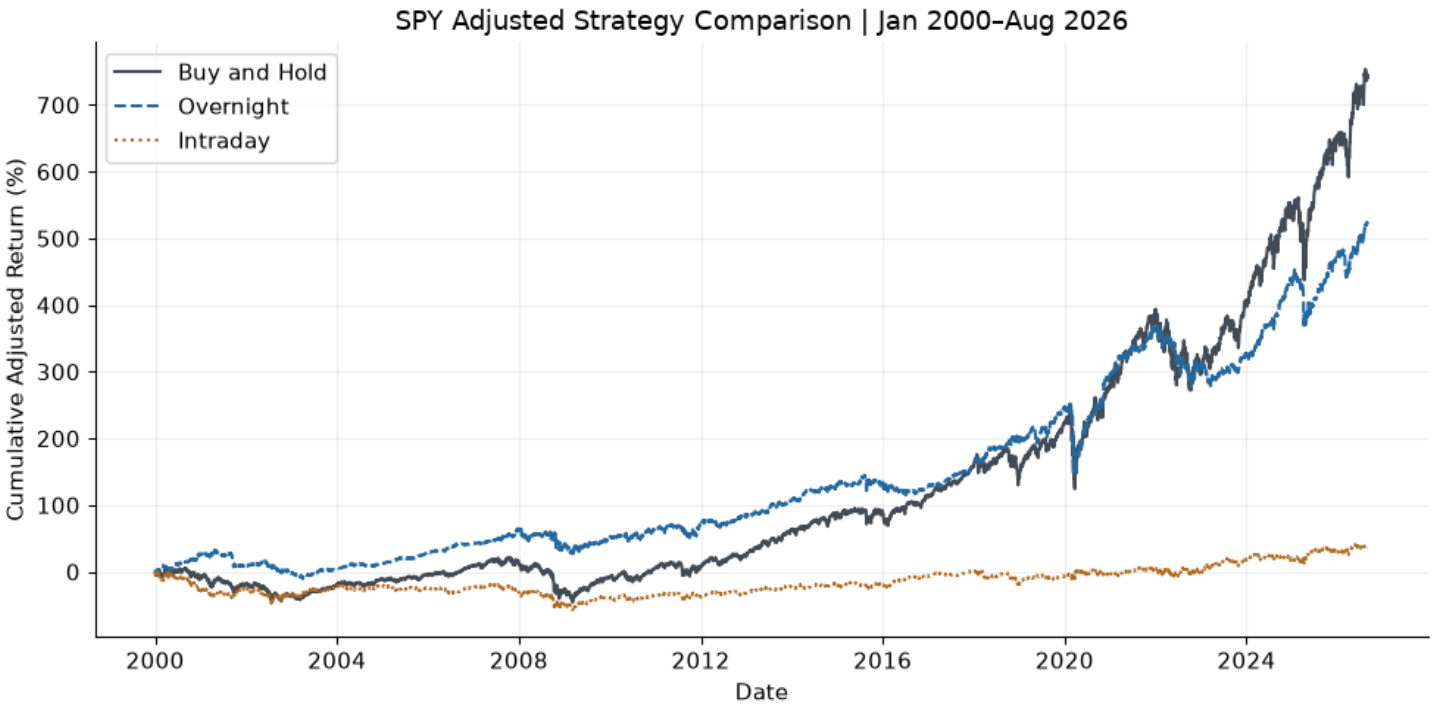
PYTHON PROJECT / RESEARCH SUMMARY

Overnight Market Anomaly

Comparing overnight and intraday ETF returns

SPY focus: Jan 2000-Aug 2026 | 6,704 daily return observations
Adjusted prices throughout; also compares QQQ, IWM, and DIA.



SPY RESULTS

Measure	Overnight-only	Buy-and-hold
Final value of $1	$6.23	$8.42
Annualized Sharpe ratio	0.67	0.51
Maximum drawdown	32.79%	55.19%

Return timing. 85.91% of SPY's cumulative log adjusted return came from overnight returns. Buy-and-hold still finished with more wealth.

Mean returns. The paired t-test did not find a statistically significant mean difference between overnight and intraday returns at 5% (p = 0.1577).

Extreme returns. Removing the top 1% of overnight observations cut growth from 6.23x to 0.94x. Those dates were identified with hindsight.

Adjusted SPY prices approximate dividend-inclusive returns; trading costs are excluded. Sharpe uses 252 trading days and a zero risk-free rate. Risk comparisons are descriptive. The test does not account for dependence across days.

Source: Yahoo Finance via yfinance | Tools: pandas, NumPy, SciPy, Matplotlib

Full analysis and code: view the GitHub repository